Advanced Programming Practice · 21CSC203P

Unit 1 · Session 5 · SLO 1 — Solved Worksheet

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Unit – 1 : Session – 5 : SLO – 1 | Classify Paradigms

Activity (Solutions)

• 1) Classification: Java → imperative, object-oriented (with functional features). Prolog → declarative, logic programming. SQL → declarative, database/query. Python → multi-paradigm: imperative, object-oriented, functional.

• 2) Feature table — Imperative: explicit steps & state changes (Java loops, Python). Declarative: describes what, not how (SQL SELECT). Logic: facts + rules + inference (Prolog parent/ancestor rules). Functional: pure functions, immutability, higher-order functions (Java streams, Python map/lambda).

• 3) Hierarchical relationship: Programming paradigms divide into Imperative (how: procedural, object-oriented, parallel) and Declarative (what: functional, logic, database/query). Each sub-paradigm refines the parent style — e.g., OOP is imperative with encapsulated state; logic and functional are declarative specializations.